

Adding Photos to wildlife-photography

How to add new photos and push them to the live site

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TL;DR

Two terminals running:

```
npm run dev           # terminal 1 - site at localhost:4321
npm run edit-captions  # terminal 2 - editor at localhost:4322
```

In the editor: **drag JPEGs onto the upload card, pick Wildlife or Misc, Claude vision auto-fills the captions, tweak as needed, save.**

Then ship:

```
npm run push-photos    # originals -> Cloudflare R2 (md5-diffed)
git add -A
git commit -m "add new photos"
git push               # Cloudflare Pages auto-deploys
```

That's the whole workflow. Live site updates within ~5–10 minutes (the first build per session is the slow one — subsequent rebuilds reuse the image cache).

Why two terminals

`npm run dev` keeps the public-facing site live at `localhost:4321`. When new files appear in `src/photography/`, the watcher regenerates `src/image-data.json` and Astro hot-reloads — so anything you add shows up in the gallery within a second.

`npm run edit-captions` opens a separate companion editor at `localhost:4322`. It is the easiest way to add new photos and edit captions: drag-drop ingestion, Claude vision auto-classify, inline edit of every field. Closing this tab is safe — your edits live in `data/animal_ids.json` either way.

What the editor does per upload

1. Saves the JPEG to `src/photography/<section>/`.
2. Runs `exiftool -gps:all=` to strip GPS coordinates (wildlife-ethics — don't post breeding-site geotags).
3. Reads EXIF for the date, dimensions, camera, lens, exposure.
4. Looks up the date in `data/locations.json`; if a matching location exists, pre-fills it on the photo.
5. If `ANTHROPIC_API_KEY` is set in `.env`, downsamples the image to 1568px and sends it to Claude Sonnet 4.6 with a JSON-schema-constrained prompt. The response — common name, scientific name, scene description, confidence, optional notes — is written to `data/animal_ids.json`.
6. Marks the file as "keep" in `data/decisions_<section>.json`.
7. Schedules a debounced rebuild of `src/image-data.json` (800 ms after the last edit). The dev server hot-reloads.

If Claude's classification is off, just click the field in the editor and rewrite it. Saves on blur. Each save re-rebuilds.

Pushing to the live site

The site lives in two places at runtime:

- **Code + metadata** in the GitHub repo (DJRGVC/wildlife-photography). This is small — JSON files, scripts, components. Commit + push.
- **Photo originals** in Cloudflare R2 (wildlife-photography-originals). R2 holds the only durable copy. Cloudflare Pages downloads them at build time, resizes them, and ships WebP variants.

After adding photos locally, both have to be updated:

```
npm run push-photos
git add -A
git commit -m "add new photos"
git push
```

`push-photos` md5-diffs against R2, so re-running is cheap and only uploads what's changed. Skipped files don't cost anything.

`git push` triggers a fresh Cloudflare Pages build that pulls from R2, processes images via sharp, and deploys the new gallery. Status visible in the **Workers & Pages** section of the Cloudflare dashboard.

When something looks wrong

Auto-classify failed during upload. The upload still saved the file with empty fields. Open the photo card in the editor and fill the fields manually. Common causes: bad or missing `ANTHROPIC_API_KEY`, transient API error, image too large after resize (very rare).

Pages build failing. Open the build log in the dashboard. Look for:

- *Missing R2 env vars.* The build can't fetch originals. Re-verify R2_ACCOUNT_ID, R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_BUCKET, R2_PREFIX, and NODE_VERSION=24 in the Pages project settings.
- *Malformed JSON in data/.* Run `npm run build:photos` locally — the same error will surface with a line number.

Dev server stopped showing new photos. Restart `npm run dev`. Astro's image-handling pipeline occasionally misses new files when the dev server was launched before the directory existed.

Removed a photo, gallery still shows it. Delete the file from disk, remove its entry from `data/animal_ids.json`, mark it "reject" in the relevant `decisions_<section>.json`, then `npm run build:photos`.

File layout, briefly

<code>src/photography/wildlife/</code>	Wildlife JPEGs, gitignored (R2 is canonical).
<code>src/photography/misc/</code>	Other JPEGs, same.
<code>data/animal_ids.json</code>	Per-photo metadata: animal, location, scene, EXIF context.
<code>data/locations.json</code>	Date → location lookup, used to pre-fill new uploads.
<code>data/decisions_wildlife.json</code>	Keep/reject flags for wildlife.
<code>data/decisions_misc.json</code>	Same for misc.
<code>src/image-data.json</code>	Generated, gitignored. Don't edit by hand.
<code>.env</code>	R2 + Cloudflare + Anthropic credentials. Gitignored. Keep private.

Live site

<https://danielgrant.photos>